

Shrikanth Vilvadrinath

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SUMMARY

AI Systems Engineer building enterprise agentic systems for lending, procure-to-pay and air-gapped regulated platforms. Fullstack **Python** (FastAPI, Flask, Django, SQLAlchemy) over SQL and NoSQL, with TypeScript front ends. Specializes in **context engineering**: **LangGraph** and **MCP orchestration**, **RAG** and **vector search**, session and context management and prompt engineering, on **scalable distributed** Temporal workflows across AWS, Docker and CI/CD.

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, Bash
Agentic AI	LangGraph, LangChain, LangSmith, MCP (Model Context Protocol), A2A, Temporal.io, Vercel AI SDK, multi-agent orchestration, tool calling, Anthropic / OpenAI / Ollama
Context Eng.	RAG pipelines, vector search, session and context management, prompt engineering, confidence thresholds, human-in-the-loop gating
Evaluation	Langfuse, agent eval harnesses, macro F1 / accuracy / QWK / safety recall, Playwright
Backend	FastAPI, Flask, Django, NestJS, Fastify, SQLAlchemy, Drizzle, REST, WebSockets
Data	PostgreSQL (RLS, multi-tenant), Supabase, Redis (NoSQL), pgvector, Qdrant
Cloud	AWS (EC2, S3, VPC, IAM), Terraform, Docker, GitHub Actions CI/CD, LiteLLM

WORK EXPERIENCE

Confer Solutions AI

Jan 2026 – Present

AI Engineer · Dallas, TX

Agentic Loan Origination Platform

LangGraph · LangChain · LangSmith · MCP · Temporal · Vercel AI SDK · Anthropic / OpenAI · Supabase · Next.js

- Designed the **context layer**: **four MCP servers** for document retrieval, mortgage data extraction and encrypted multi-party email, behind a **LiteLLM gateway** that keeps every agent portable across Anthropic, OpenAI and local models.
- Built the **work assignment engine** routing loans to processors across four strategies: an **O(1)** round-robin pointer keyed on (role, organization), and load-balanced routing that ranks live active-loan counts under a per-user workload cap.
- Set the **autonomy boundary** across **four independently signaled Temporal phase workflows**: three blocking human gates over 12 lifecycle gates, auto-clear only above **0.90 confidence**, and stall detection that scopes a failure to one phase.

Change Management & Release Governance – IL5/IL6 Air-Gapped

Terraform · AWS (VPC, EC2, IAM) · Langfuse · Temporal · Docker · Gitea · Ollama · Next.js

- Provisioned the **AWS** environment in layered **Terraform** under a fixed IAM permission boundary (VPC, EC2 with **IMDSv2**, encrypted EBS), then built the release pipeline as a **9-step Temporal workflow** – preflight, Zarf packaging, cosign signing, OCI push, migration, deploy, health verification and drift scan – moving signed **163 MB** images into an enclave with no egress.
- Orchestrated **five agents in parallel** (change risk, compliance, process drift, issue intelligence, release advisor) and built the **evaluation harness** they are scored on: macro F1, QWK and safety recall over labelled examples, traced to a self-hosted **Langfuse**. Verified the release advisor at **87.5% accuracy with zero false-ready calls**.

Procure-to-Pay ERP

TypeScript · NestJS · Temporal · PostgreSQL (RLS) · Drizzle · Docling · Docker · GitHub Actions

- Engineered **one signal-driven Temporal approval ladder**, assembled at runtime from an approval-matrix table, that serves all **33 business flows** with **maker-checker**, **N-of-M quorum** and **24/48/72-hour SLA** escalation over a multi-tenant Postgres carrying **80 row-level-security policies**. New approval rules ship as configuration, with no redeploy.
- Built the **document intelligence pipeline**: a photographed goods-receipt note is parsed on CPU in **~98 s** and matched to its purchase order across a **15-field** schema. Anything under **0.75 confidence** routes to review, so no extraction reaches the ledger unverified.

NV Rad Imaging

May 2025 – Dec 2025

AI Engineer · Tampa, FL

Python · LangGraph · LangChain · Ollama · Supabase · Next.js

- Built an **agentic patient intake platform**: agents extract and structure submitted forms, run eligibility checks and send follow-ups with **no manual handoff** between stages.
- Shipped a clinical **RAG question-answering pipeline** over indexed diagnostic reports, cutting review time **~35%** and resolving **90%+ of queries autonomously** across **100+ reports**.

PROJECTS

Git Gardener – Agentic Repository Intelligence

gardener.twoby2.dev

Python · FastAPI · SQLAlchemy · PostgreSQL · Temporal.io · Next.js · Docker

- Fullstack Python service that decomposes any GitHub repository into a **Temporal activity chain** with per-step retries, classifying files, scoring complexity and producing an LLM-generated cleanup roadmap ranked by impact, persisted through SQLAlchemy.

The Pantheon – Autonomous A2A Newsroom

pantheon.twoby2.dev

Google ADK (A2A) · Python · Next.js · Supabase · DigitalOcean

- Eleven autonomous agent roles** across two teams, one running always-on services and one shipping code, coordinate with no human trigger by brokering task envelopes over the **A2A protocol** into one shared, auditable schema.

EDUCATION

University of Maryland, College Park – MS, Data Science · GPA 4.0

Aug 2024 – May 2026

IIT Madras – BSc, Data Science & Applications · GPA 3.5

Jan 2021 – Dec 2024

Vishwakarma Government Engineering College – BTech, Electronics & Communication · GPA 3.5

Sep 2020 – Jun 2024